

Problem 8

John starts with \$2 and $p = 0.6$. What is the probability that John wins if the opponent has \$5?

Solution:

$$P_2 = \frac{1 - \left(\frac{0.4}{0.6}\right)^2}{1 - \left(\frac{0.4}{0.6}\right)^7} = 0.59$$

Problem 9

John starts with \$3 and $p = 0.5$. What is the probability that John wins if the opponent has \$5?

Solution:

$$P_3 = \frac{3}{N} = \frac{3}{10} = 0.3$$

Problem 10

John starts with \$2 and $p = 0.6$. What is the probability that John will become infinitely rich if the opponent has infinite capital?

Solution:

$$\lim_{N \rightarrow \infty} P_2 = 1 - \left(\frac{q}{p}\right)^2 = 1 - \left(\frac{0.4}{0.6}\right)^2 = 0.56$$

Problem 11

John starts with \$2 and $p = 0.5$. What is the probability that John will become infinitely rich if the opponent has infinite capital?

Solution:

$$\lim_{N \rightarrow \infty} P_2 = 0$$

Problem 12

You bought a share of stock for \$12. Stock price moves \$1 each day as a simple random walk. The probability of an increase is 0.52.

- i) What is the probability that the stock price will reach \$16 before it reaches \$9?
- ii) What is the probability that you will get infinitely rich?

Solution:

i)

We want the probability that the stock price goes up by 4 before going down by 3. That is, $a = 4$ and $b = 3$.

$$p(4) = \frac{1 - \left(\frac{0.48}{0.52}\right)^3}{1 - \left(\frac{0.48}{0.52}\right)^7}$$

ii)

We use $i = 12$ in the gambler's ruin problem:

$$\lim_{N \rightarrow \infty} P_{12} = 1 - \left(\frac{q}{p}\right)^{12} = 1 - \left(\frac{0.48}{0.52}\right)^{12}$$